

Improving Reading Instruction and Students' Reading Skills in the Early Grades: Evidence From a Randomized Evaluation in Haiti

Juan Carlos Guzmán

Kate Schuenke-Lucien

Anthony J. D'Agostino

Mark Berends

University of Notre Dame, Indiana, USA

Andrew J. Elliot

University of Rochester, New York, USA

ABSTRACT

In Haiti, 49% of students cannot read a single word in Creole by the time they start grade 3, which is reflective of a broader learning crisis in low-income and fragile contexts. Read to Learn, an early-grade literacy intervention, was implemented and evaluated from fall 2014 through spring 2016 with the aim of improving students' reading skills. Students were given learning materials in their mother tongue, teachers were provided with training and instructional coaching, and various supports for program implementation were established. In a randomized evaluation, the authors assessed students' reading skills at the beginning of grade 1 and at the end of grades 1 and 2 in treatment and control schools. The authors estimated the impact of the program at the end of grades 1 and 2 with a hierarchical linear model and found positive effects on emergent reading skills and oral reading fluency, with effect sizes ranging between 0.19 and 0.79. The results of this study are an important contribution to knowledge about what works to improve literacy outcomes for students in Haiti and other fragile contexts.

Education is of critical importance in the poorest countries in the world. It is a path for children to pursue lives of dignity and opportunity amid poverty. The global community has recognized the importance of education over the past 50 years and made universal primary education a goal. Both Education for All and the Millennium Development Goals promote universal access to primary education (Chabbot, 2014). The goal of universal access has been implemented in national policies throughout the developing world. As a result, the number of out-of-school youth of primary school age was reduced by half in 2011 (United Nations, 2013).

It soon became clear that access to school was not enough to improve student learning outcomes. Students need access to high-quality education within schools to learn. Variance in personal and economic growth influences learning outcomes (Hanushek & Kimko, 2000; Hanushek & Woessmann, 2008; Pritchett, 2013). Researchers examined how much students in low-income countries were actually learning and concluded that a large segment of students was not achieving basic skills in subjects such as reading (Gove & Cvelich, 2011). This result has shifted the focus from access to education to quality of education over the past 15 years (Pritchett, 2013; Wagner, 2017). As a result, greater numbers of scalable and sustainable programs to improve students' basic skills are being implemented in low-resource contexts. Early-grade reading is a foundational skill for all future learning and has received much attention (Kim, Boyle, Zuilkowski, & Nakamura, 2016).

We examined the impact of Read to Learn, a systematic reading intervention implemented in Catholic schools in Haiti, on reading skills in grades 1 and 2. In Read to Learn, students learn to read in Haitian Creole first and then receive literacy instruction in French. The results of our study contribute to knowledge about what works in education interventions in fragile, low-resource contexts. The results also add to the literature on the impact of scripted curricula on reading skills. In this article, we describe the context of the Read to Learn program and present the methods and results of our analysis. We conclude with a discussion of the findings and offer suggestions for further research.

The Haitian Context

Haiti occupies one third of the island of Hispaniola in the Caribbean. It is the poorest country in the Western Hemisphere and one of the poorest in the developing world. Haiti's gross domestic product per capita was \$870 in 2018. It had a human development index ranking of 163 out of 188 countries in 2016 (World Bank, 2019). Over 6 million Haitians live on less than \$2.41 per day, and over 2.5 million Haitians live below the extreme poverty line of \$1.23 per day (World Bank, 2019). In Haiti, 22% of children are stunted and enter school malnourished (UNICEF, 2016).

Over 95% of Haitians speak Creole as their first language. French is the traditional language of reading instruction, but most teachers and students speak it with low or no fluency (Gönsch, 2010). No more than 5% of Haitians speak French, and only 5–10% of the population has receptive competence in the language (Foster & Valdman, 1984; Zéphir, 2015). Among Haitian teachers, only 20% speak French fluently (Hebblethwaite, 2012). This lack of fluency creates a “pedagogical, sociocultural, economic, and political predicament” (DeGraff, 2016, p. 436) common in Haiti and other postcolonial societies. The memorization of French texts exacerbates issues stemming from the language of instruction (DeGraff, 2016; Jean-Pierre, 2015). Literacy instruction often relies on the “memorization of letter-to-sound correspondences, with students ‘reading’ texts that they often do not understand” (DeGraff, 2016, p. 437). This approach leads to poor results both in literacy skills and in future learning outcomes (Messaoud-Galusi & Miksic, 2010).

Unlike French, Haitian Creole does not use an opaque orthography. Haitian Creole is a transparent language with a completely distinct grammatical structure from French (Dejean, 1986). A simplified explanation for how Haitian Creole came to be is that it emerged from French molded into the African syntax of the Gbe language of the Kwe family. Haitian Creole has phonological representation like French expression but lexically shares properties

with African substratum languages. French is the superstratum language of Haitian Creole, but the typology between the two are different (DeGraff, 1992). Although 90% of Haitian Creole vocabulary is of French origin, the two languages are jointly incomprehensible. For effective literacy instruction, teachers must ensure that students understand that French and Haitian Creole are two distinct languages.

Due to low-quality education, students in Haiti are unable to achieve basic competence in literacy and numeracy (Ball, Paris, & Govinda, 2014; Nielsen, 2014). When they enter third grade, 49% of Haitian students are not able to read a word in Creole, and 48% cannot read a word in French (RTI International, 2012). A shortage of qualified teachers hampers the school system. Almost 80% of Haitian teachers have minimal preservice and reading instruction training (USAID, 2016). Poor quality of instruction is a top reason for Haitian students who drop out of school early (Crouch, 2005; World Bank, 2007). Haiti suffers from a persistent and pervasive school completion crisis. A minority of students completes primary school, and even fewer complete secondary school (Dejean, 2006). Only 10% of students who enter first grade go on to complete secondary school (Groupe de Travail sur L'Éducation et la Formation [Education and Training Working Group], 2010). Under-enrollment among school-age children and limited government oversight exacerbate Haiti's educational situation (USAID, 2016).

In Haiti, limited evidence exists about effective methods for improving literacy skills in the early grades. Save the Children conducted a baseline–endline evaluation of an unscripted literacy intervention, *Lekti se Lavni* (Reading Is the Future), in Haitian Creole in 36 schools from October 2012 to June 2013. This intervention took place in schools in Port-au-Prince and Leogane (the West Department) and included “teacher training, community reading activities, and the creation of age-appropriate local language material to support emergent literacy skills among early-grade children (kindergarten to second grade)” (Save the Children, 2013, p. 3). Save the Children did not reveal whether the 36 schools in the study were public, private, or a combination, simply noting that they were supported by Save the Children. Save the Children found gains in student accuracy (number of words read and understood) but observed very limited gains in reading comprehension.

RTI International conducted a randomized evaluation of a scripted literacy program in 160 treatment schools and 80 control schools in the North, West, and Artibonite departments (regions) of Haiti between 2012 and 2014. In a mix of public and private schools, Haitian teachers were equipped with a scripted Creole and French literacy curriculum called *Tout Timoun Ap Li* (All Children Reading; ToTAL). RTI International developed ToTAL in collaboration with the Haitian government and with

funding from USAID from 2011 to 2013. The overarching goal for ToTAL is to improve reading skills for early elementary students in Haiti (for further details of the program, see RTI International, 2015). In the ToTAL curriculum, students use reading and writing books in Haitian Creole in first grade that are aligned with teacher guides in Haitian Creole, and they also use picture books that align with teacher guides in oral French. It is assumed that students enter first grade fluent in Haitian Creole as their mother tongue and have limited proficiency in French. The goal of ToTAL in first grade is to start teaching students to read in Haitian Creole, with a focus on introducing letter names, letter sounds, and phonemic awareness. In grade 1, the ToTAL curriculum also focuses on teaching French orally to increase students' understanding, allowing them to begin reading instruction in French in second grade. In grade 2, the ToTAL curriculum includes reading and writing books in both Haitian Creole and French that are aligned with teacher guides, with the goal of building on the reading skills achieved in Haitian Creole in first grade and beginning to teach reading skills in French. In the second-grade curriculum, students continue to develop their French-language skills while moving toward blending letter sounds, decoding, reading initial words, and achieving oral reading fluency in both Creole and French.

The RTI project resulted in strong gains in initial sound identification, substantial gains in the identification of letter sounds, and promising trends in listening comprehension (RTI International, 2015). RTI International did not find positive effects in other skill areas measured. RTI International attributed these results to low implementation fidelity and acknowledged that many teachers did not actually receive the materials needed to deliver the curriculum to students or did not use the teaching materials in the correct manner (RTI International, 2015).

These two studies on *Lekti se Lavni* and ToTAL resulted in the publication of project reports, but they were not peer-reviewed studies. To our knowledge, no published peer-reviewed studies of literacy interventions in Haiti exist.

Scripted Literacy Interventions as a Solution to Haiti's Literacy Crisis

New evidence about what works to improve literacy indicates that scripted programs may be effective for teaching reading in contexts such as Haiti. Promising approaches involve elaborated teaching materials, student curricula, and professional development for teachers (Snilstveit et al., 2016). Highly structured or scripted approaches to literacy instruction in developing countries with similar teacher

and student profiles as those in Haiti have shown great success (Davidson & Hobbs, 2013). A meta-analysis of 216 programs reaching 16 million students found that although there are no magic bullets, structured pedagogy programs have consistent positive effects on students' reading and mathematics achievement, with average effect sizes of 0.23 and 0.14, respectively (Snilstveit et al., 2016). Research has indicated that structured literacy programs are most effective when they are tailored to the linguistic context in which they are being implemented, such as teaching in the language spoken at home and including materials in the mother tongue (Ball, 2010; Bühmann & Trudell, 2008; Cummins, 2000). Some of the smallest effects have been observed for programs taught in students' second language (Lucas, McEwan, Ngware, & Oketch, 2014; Spratt, Ralaingita, Fomba, & Diawara, 2013). Additionally, interventions that only provide text resources without instructional supports have shown limited or no effects on improved learning outcomes (Yi et al., 2019).

When using scripted curricula, positive effects have been reported for students' reading skills in high-income countries, including the United States (WWC, 2017a). For example, *Success for All* provides scripted reading materials, software, and extensive training to participating teachers (Cheung & Slavin, 2016; Slavin, Madden, Chambers, & Haxby, 2009). Evaluations of *Success for All* found positive effects on phonics and reading fluency but mixed effects on reading comprehension (Borman et al., 2007; Cheung & Slavin, 2016; Cohen, Peurach, Glazer, Gates, & Goldin, 2014; Peurach, 2011; Quint et al., 2014; Rowan, Correnti, Miller, & Camburn, 2009).

Despite the evidence on scripted curricula, they are not without controversy (Snilstveit et al., 2016). The effects of these programs on reading comprehension are weak. The programs have been criticized for an overemphasis on phonics and decoding and an underemphasis on reading practice and discussion to make meaning from text (Abadzi, 2006; Bartlett, Dowd, & Jonason, 2015; for critiques, see Pogrow, 1998; Walberg, 1998).

In Latin America, there are examples of other successful approaches to improving literacy skills at scale, including Freire's generative word method (Monteith, 1977), Cuba's *Each One Teach One* (Moody, 2018), and the syllabic approach employed in Mexico (Reese, Mejía-Arauz, & Bazán, 2012). These approaches to literacy instruction require human capital in the form of basic literacy skills within the general population or a literate, professionalized, and stable teaching force. None of these conditions are a current reality in Haiti. Haiti's socioeconomic conditions are closer those present in sub-Saharan African nations, many of which have used scripted, mother-tongue literacy programs with success (Piper & Korda, 2011; Tilson, Kamlongera, Pucilowski, & Nampota, 2013).

The 5Ts Framework

One advantage of scripted approaches to literacy, such as the approach used in Read to Learn, is that they allow for the incorporation of elements commonly associated with effective literacy interventions in contexts of poverty. The 5Ts of program design and implementation that have been shown to have a positive impact on literacy outcomes globally are teaching, time, texts, tongue, and test (Kim et al., 2016).

Teaching

The teaching element refers to adequate training and support for teachers. Programs that provide supplemental material, follow-up visits, and a focus on attending to specific school constraints are associated with gains in student learning (Piper, Zuilkowski, & Mugenda, 2014). Effective programs promote scaffolded instruction and a gradual release of responsibility (Duke & Pearson, 2002; Pearson & Gallagher, 1983). They often adopt some form of the “I do, we do, you do” model (Kim et al., 2016). Teachers use scripted guides to help prepare lessons and integrate principles of effective literacy instruction (Tilson et al., 2013). Research has pointed to the need for rigorous professional development to accompany the use of scripted lessons (Allan & Horn, 2013).

Time

This element refers to the importance of instructional time for effective literacy outcomes (Levin & Lockheed, 1993). Literacy is often not taught as a separate subject or allocated enough time in fragile contexts (Bunyi, Cherotich, & Piper, 2012; Gilles & Quijada, 2008). Studies in Haiti found that schools closed 27% of instructional days and that students attended classes only 77% of the time (DeStefano & Miksic, 2007). Gilles and Quijada (2008) showed that lost instructional time explained 55% of the variation in Haitian students’ reading skills.

Texts

This element refers to the availability of texts that are appropriate regarding students’ age, ability-level, and language. Texts and reading practice are critical for developing students’ automaticity (Abadzi, 2006; Breznitz, 2006; Ehri, 2005). Texts and reading practice also contribute to developing reading comprehension and vocabulary (Bartlett et al., 2015; Hayes & Ahrens, 1988).

Tongue

This element points to the importance of reading instruction in students’ mother tongue. Such instruction includes teaching in the local language and materials available in the mother tongue (Ball, 2010; Bühmann & Trudell, 2008;

Cummins, 2000). Programs taught in students’ second language have shown the smallest effects (Lucas et al., 2014; Spratt et al., 2013). Studies of literacy acquisition in Kenya, a multilingual context, have indicated that mother-tongue instruction may improve students’ early literacy skills in both their mother tongue and their second language (Piper, Schroeder, & Trudell, 2016).

Test

The test element pertains to various forms of assessment that programs can and should integrate. Students are seldom tested in low-income countries. Tests occur at transitions in the schooling level (e.g., from elementary to lower secondary). Thus, data are not available to improve instruction at the classroom and school levels (Kim et al., 2016; Learning Metrics Task Force, 2014). Classroom-level assessments matter most for informing teachers about individual student learning progress and allowing them to adjust instruction accordingly (Fuchs & Fuchs, 1999, 2002).

The Current Study

The research question guiding this study was whether the Read to Learn program had an impact on the literacy skills of Haitian students. We employed a randomized controlled trial to evaluate students’ reading results in both treatment and control schools to address this question. We conducted three rounds of data collection to measure the impact of the study. The first round took place before the intervention began at the beginning of first grade. The other two rounds occurred at the end of grades 1 and 2. Because there were different levels of student and school attrition, we conducted separate analyses for each grade: one with a sample of students whom we were able to follow and test at the beginning and end of first grade and another with a sample of students whom we were able to follow and test at the beginning of first grade and at the end of second grade.

Method

Participants

A total of 1,189 students who began first grade (632 students in control schools and 557 students in treatment schools; 50% female; mean age = 5.9 years) were randomly selected from 94 Catholic schools in the regions of Gonaïves (732 students) and Cap-Haïtien (457 students) in Haiti (see Table 1 for the demographics and test scores). All students were of Haitian descent and came from homes where Haitian Creole is the dominant language.

TABLE 1
Participant Characteristics and Test Levels at Baseline (N = 1,189)

Measure	Control (n = 632)		Treatment (n = 557)		t	p
	M	SD	M	SD		
Participant characteristics						
Sex (percentage female)	50.8	50.0	50.1	50.0	0.24	.81
Age (years)	5.56	1.36	5.68	1.37	-1.53	.13
Average first-grade class size	38.70	12.83	36.10	13.40	3.42	<.01
Region (percentage Cap Haitian)	58.2	49.4	65.4	487.6	-2.53	.01
Early Grade Reading Assessment at baseline						
Letters read per minute	9.05	13.95	10.67	15.87	-1.86	.06
Letter sounds pronounced per minute	7.75	11.82	9.77	18.23	-2.24	.03
Words read per minute	4.01	8.49	4.84	14.08	-1.20	.23
Invented words read per minute	3.64	8.98	4.06	11.12	-0.71	.48
Words in a story read per minute	7.84	18.79	8.34	22.30	-0.42	.68
Initial sound recognition	18.23	30.92	21.10	34.00	-1.51	0.13

Note. M = mean; SD = standard deviation.

Sampling Procedures

In collaboration with the Haitian Episcopal Commission for Catholic Education, we established criteria for schools to participate in the evaluation of Read to Learn. Schools were eligible if they were Catholic, belonged to one of the regions, and met the following four criteria: in need of strengthening, a basic level of functionality, minimally competent staff (i.e., teachers who could read), and interest in participating in the program. Accordingly, we randomly assigned schools to the treatment and control groups by sorting random values and assigning the first half of the sorted list to treatment and the second half to control within each region.

In each school, we randomly sampled students to be tested at the beginning of first grade for the baseline sample. We used a random number generator app that selected 20 numbers, from 1 to the total number of students in first grade, without replacement. This list of random numbers was not in order and determined which students were selected according to their positions on the first-grade roster. Students absent from school the day of the test were excluded from the roster list prior to matching the list of random numbers to students' names. Students were selected from all first-grade classrooms if there were multiple classrooms (26 schools had more than one classroom). The probability of selection varied per school from 100% for five schools to 6%. No student not chosen to be tested was accepted into the sample.

The schools were visited three times by trained enumerators. Data and students' test results were collected using Android tablets. We used the Tangerine application designed by RTI specifically to collect the reading test results. The enumerators were trained for each visit in a three-day workshop on how to use the tablet and how to administer the test. One or two enumerators visited each school and requested all of the first-grade student rosters. The schools received previous notice of this evaluation by the Haitian Episcopal Commission for Catholic Education. The enumerators tested one student at a time in a private space provided by the school.

This study was reviewed and approved by the Institutional Review Board at the University of Notre Dame and by the Haitian Episcopal Commission for Catholic Education board.

Sample Size and Power

We selected a total of 94 schools to be part of the evaluation: 49 schools were selected to be part of the control group, and 45 schools were selected to be part of the treatment group. The sample size per school was determined by a power calculation for two-level cluster designs to achieve an effect size of 0.2 with 80% power and a 5% significance level. We also increased the result of the power calculation by 20%, as we assumed the same level of attrition at the student level. We attempted to visit and test the same sample of students in two follow-ups. The sample

sizes for these follow-ups were different than that for the baseline sample. The magnitude and implications of these differences are discussed in the Results section.

Intervention

As previously discussed, Catholic schools in the North and Artibonite departments in Haiti were randomly assigned to the treatment and control groups. The Catholic school system was a reasonable context in which to test Read to Learn. The overwhelming majority of Haitian students, approximately 80–90%, are enrolled in private schools (RTI International, 2015). Catholic schools are the largest provider of educational services in Haiti. The Catholic school network comprises 17% of private schools (Ministère de l'Éducation Nationale et de la Formation Professionnelle [Ministry of National Education and Vocational Training], n.d.).

The control schools continued to use the standard literacy materials available in Haiti for teaching Creole and French to first- and second-grade students: Mwen Konn Li and Joupi (I Know How to Read, Yippee). It is important to note that Creole is supposed to be the language of instruction in early grades in Haiti. The Bernard reform of 1982 in Haiti mandated Creole as the medium of instruction in the early grades, but this decision was met with broad social resistance. Thus, even though Creole is the legally mandated language of instruction in early primary grades in Haiti, teachers and school administrators prioritize French, and most teachers receive little or no formal training in teaching students to read in Creole (Gibson, 2011). Notably, the general curriculum used in the control schools allocates less time to literacy instruction than the Read to Learn program requires. The control schools followed the national curricular requirements of 30 minutes three days a week dedicated to literacy instruction in both French and Creole. The materials used in the control schools (traditional curriculum) were 60–75-page lesson books with extremely limited opportunities for meaningful interaction between teachers and students. These lesson books were designed for cost-effectiveness; the curriculum distributor in Haiti must sell books that are inexpensive enough that parents can afford to purchase them from local vendors, so the result is very thin lesson books. Based on observational visits to control schools by program staff, teachers often assign students to simply copy elements of the lesson onto the classroom chalkboard or to memorize the lessons in choral repetition. Teachers in control schools continued to receive the standard coaching visits from their respective Diocesan Offices of Education that are periodically provided to all Catholic schools in these dioceses in Haiti. Teachers and school directors in the control schools did not receive any special literacy coaching or professional development.

The treatment schools participated in the Read to Learn intervention, which was implemented by the University of Notre Dame, the Haitian Episcopal Commission for Catholic Education (equivalent to the National Office of Catholic Schools for Haiti, with ultimate authority over all 2,400+ Haitian Catholic schools), and Catholic Relief Services. Read to Learn incorporated the ToTAL curriculum developed by RTI, which is a scripted early-grades literacy program in students' mother tongue, along with robust coaching and support systems for teachers. Through the design of the Read to Learn program, the partners in the Catholic sector addressed two major challenges of literacy instruction in Haiti: untrained teachers and inadequate curricular materials available for literacy instruction in Creole (students' mother tongue) and in French (students' second language). The students in the treatment schools received the ToTAL student and teacher materials, which included reading and exercise books in Creole in grades 1 and 2 with aligned teacher guides, an oral French teacher guide with an aligned student picture book in grade 1, and student reading and writing books in French with an aligned teacher guide in grade 2. Instructional posters and basic school supplies (chalk, pencils, and erasers) were given to the Read to Learn classrooms.

The Read to Learn partners sought to apply lessons learned from the RTI literacy intervention from 2012 to 2014, which suffered from low implementation fidelity, to improve the impact of the program. The Haitian Episcopal Commission for Catholic Education had worked as a contractor under RTI from 2012 to 2014, and its pedagogical staff was convinced that the ToTAL curricular materials were of high quality and could be effective if they were used correctly in classrooms. Read to Learn used the same curriculum as the RTI intervention (the ToTAL curriculum) but also focused on incorporating the 5Ts (i.e., teaching, time, tongue, texts, test) into the intervention design, with varying degrees of success. For the teaching element, teachers received a week of residential pedagogical training. The training focused on the theory behind structured literacy teaching techniques. Teachers also practiced lessons with support and feedback from other teachers and trainers. Once teachers began giving lessons, they received ongoing support in the form of bimonthly visits from coaches. Coaches had training in the use of the curriculum and experience as teachers in using the ToTAL curriculum. During their visits, they focused on observation and instructional coaching. The Read to Learn project also included training for school directors aimed at increasing school-level buy-in and school-based support for teachers as they implemented the program. Using teacher's guides, teachers were taught to use scaffolding instruction with a gradual release of responsibility ("I do, we do, you do"; Kim et al., 2016). The guides had a spiraled design with frequent opportunities to allow students

to review and master previous content. If teachers followed the guides, they could guide practice, check for understanding, and provide feedback.

Regarding the time element, Read to Learn required between 60 and 90 minutes of instruction per day to complete both the Creole and French lessons. In first grade, the Creole lessons ranged between 45 and 60 minutes, and the oral French lessons ranged between 20 and 30 minutes. In second grade, the Creole and French lessons were of comparable length, each approximately 45 minutes. For the texts element, Read to Learn included stories in the student reading books in Creole in grades 1 and 2 and in French in grade 2. The Read to Learn classrooms also received library kits that included 40 fiction and non-fiction storybooks in Haitian Creole and French, which teachers were instructed to read aloud with students. Teachers were also instructed to give children access to these texts for free-reading time before and after school, during the recreation period, and during class. The implementers of Read to Learn also attended to the tongue element, providing textbooks, library books, and teacher's guides in both Haitian Creole, the mother tongue of 95% of the students, and French for grades 1 and 2. In first grade, teachers used the curriculum that emphasized emergent literacy skills in Creole and introduced oral French. In second grade, teachers used the curriculum that continued the development of Creole reading and writing skills while also focusing on decoding and building fluency in Creole and French.

In addition, Read to Learn attempted to include a focus on the test element by including formative assessments for

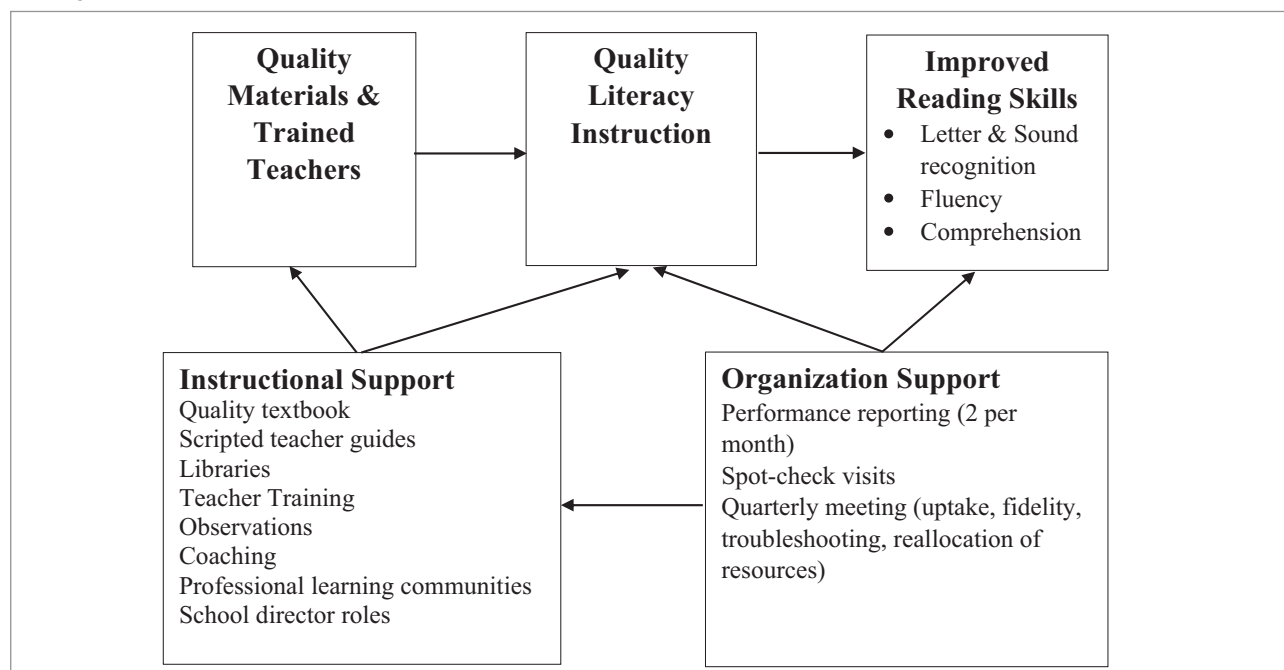
teachers in their weekly lessons. As further elaborated later, teachers did not widely use these assessments. To fill this testing gap, Read to Learn relied on informal assessments performed by coaches during their classroom visits. These assessments often consisted of reviewing students' work in their exercise books. Coaches used this strategy with teacher practice to determine which teachers needed extra coaching visits.

Theory of Action

A theory of action refers to how an intervention is delivered and implemented and how it promotes positive student outcomes. Evaluation of previous programs in Haiti suggested that implementation fidelity had been a challenge (RTI International, 2015). As such, the project placed a substantial emphasis on supporting effective implementation.

Improving students' literacy depended on teachers using the materials and delivering the lessons. Coach visits helped ensure implementation fidelity through observation and instructional coaching. These actions refined teacher practice and enhanced the efficacy of delivery. All of these aspects connected teachers' literacy instruction with students' reading skills (see Figure 1). The program staff encouraged teachers to deliver the lessons with fidelity, which involved teaching a full Creole and a full French lesson daily, delivered as close as possible to the script provided in the teacher's guides. Coaches and school directors attempted to track teachers' performance on an observation form, on which coaches recorded information on the

FIGURE 1
Theory of Action for Read to Learn



number of completed lessons, whether teachers were following the script as written, and whether they were able to complete a full lesson in the time allotted in the school day. A lack of internet access and electricity hindered reliable data collection from coaches who received handheld tablets with the observation forms to use when they observed teachers. Despite difficulties with using the tablets, the partners did not fully utilize a paper-and-pencil method of data collection as an alternative because local staff was not available to enter the paper-and-pencil results into an electronic system and because the partners believed that the issues with the uploading of data from the handheld tablets could eventually be resolved. Some information was collected in paper-and-pencil format and shared but was never entered in an electronic form. Unfortunately, these issues were never fully resolved, which limited the amount of data available for understanding why and how Read to Learn worked (or did not work).

Fidelity in Implementation

As shown in Figure 1, organizational support facilitated the implementation of literacy instruction. As noted earlier, data on implementation fidelity were not collected and shared consistently for the duration of Read to Learn. The partners collected snapshots of what was happening in the classrooms, as some months of observational data were uploaded, and some paper-and-pencil forms and the direct experiences of coaches in classrooms were shared during quarterly meetings with all partners between 2014 and 2016. It would have been ideal to have continuous observational data from classrooms throughout the program. Because we were not able to capture those data, we were forced to explain levels of implementation fidelity using the limited data we have, while acknowledging that this issue should be improved in future iterations of programs such as Read to Learn.

Coaches provided instructional coaching visits to maximize accountability and support. School directors also received training in the literacy program components because they needed to support and participate in the monitoring and coaching in their schools. Regional staff and coaches encouraged school directors to conduct weekly 15-minute observational visits to teachers and provide feedback after each visit. Oral check-in reports were provided monthly between the Diocesan Board of Education staff who oversaw the work of the Read to Learn coaches and the local Catholic Relief Service project managers. In the check-in calls, the partners were informed that coaches visited the schools at least once per month. Coaches visited many schools twice or more per month according to these reports, which also indicated that roads affected by inclement weather and political demonstrations often prevented coaches' visits. Again, the original (and ideal) plan was to have proof of these

visits based on the uploaded data from the coaches' handheld tablets (which would have included time stamps and GPS coordinates of the observations). In the absence of those data, the partners relied on the more informal feedback received from partners.

In addition, the implementing partner conducted monitoring visits to schools to understand implementation challenges, which included teachers not completing the entire lessons or falling behind in the pace of delivering the curriculum. Many teachers struggled to deliver the new lessons in the allotted time. Reasons for this difficulty include their failure to familiarize themselves with the lessons before teaching, their desire to use the old literacy materials they had in their classrooms alongside the new materials, and their perception that the new program was too burdensome. Teachers also generally failed to utilize the classroom libraries; coaches noted in partner meetings (which happened on a quarterly basis throughout the project) that the classroom library books often remained locked in the school director's office because of fear of the books being lost or damaged. The partners reassigned coaches from competent schools to struggling schools. Thus, coaches conducted more monthly visits to schools where teachers were struggling. In these cases, coaches emphasized the completion of the most basic elements of the program: Teachers should complete the full lesson in both Creole and French every day according to the curriculum.

Measures

We measured reading skills with the Early Grade Reading Assessment (EGRA), which was derived from the Dynamic Indicators of Basic Early Literacy Skills assessment (Dubeck & Gove, 2015; Good & Kaminski, 2002). The EGRA is designed to measure foundational literacy skills that individuals need for beginning reading. International educators and reading experts provided input on the development of EGRA, which has now been used in over 65 countries and in over 100 languages (Gove & Wetterberg, 2011). The theoretical underpinnings of the EGRA were informed by research describing the skills that can be reliably measured to predict initial and later reading acquisition (Dubeck & Gove, 2015). There has also been research to establish the criterion validity of oral reading fluency assessment and achievement tests, with correlations ranging from .49 to .83, indicating that oral reading fluency is a valid proxy for literacy skills (Piper et al., 2014).

In 2009, RTI International adapted the EGRA to the Haitian context and conducted reliability testing. In a 2010 report to the Haitian Ministry of Education, RTI International reported strong internal reliability for both French and Creole based on Cronbach's alphas that were well above the generally acceptable level of .70, with .91 for French and .92 for Creole. These results did not vary

when each section was eliminated, except for listening comprehension. In this case, the Cronbach's alpha increased slightly, indicating that this task contributed less strongly to the internal reliability of the overall instrument. However, the precise numbers for the EGRA subtasks were not published (Messaoud-Galusi & Miksic, 2010). The Cronbach's alpha for each subsection in the data used for this study was .97 for reading letter names, .92 for sound identification, .98 for letter sound pronunciation, .96 for word reading, .97 for invented word reading, .96 for words read in a paragraph, .72 for reading comprehension questions, .68 for Creole listening comprehension, and .78 for French listening comprehension.

In each of the three administrations of the EGRA, the administrators, who were not involved in the program implementation and were blind to the assignment of treatment and control schools, tested each student individually in a one-on-one interview in a separate room of the school. Each test administrator used the Tangerine application to record students' answers. The test administrators received a three-day training before the first round of data collection. The test administrator for the baseline assessment received a two-day refresher course on EGRA administration before collecting data for each of the two follow-ups. The test administrators learned and practiced

response collection for all EGRA subsections in groups and pairs and also received training in using the Tangerine application and the Android devices. Finally, the test administrators practiced with students from a school not included in the sample.

The three EGRA administrations comprised the following subtasks provided in Haitian Creole: knowledge of letter names, initial sound identification, knowledge of letter sounds, familiar word reading, invented word decoding (students read nonsense words that they were forced to decode), oral reading fluency (how many words in a paragraph students could read), reading comprehension (only certain students were administered this measure, not the full sample), and listening comprehension. There was also a listening comprehension subtask in French. Table 2 presents the full set of subtasks and the specific indicators used in our research.

We obtained other measures, such as the number of students in first grade and the number of classrooms for first grade, from the Diocesan Offices of Education to calculate the average class size. We also attempted to fully document the coaches' visits, but unfortunately, as noted earlier, these data were not collected systematically. Because of their poor quality, these data were not included for statistical analysis.

TABLE 2
Description of the Subtasks and Indicators of the Early Grade Reading Assessment

Subtask	Objective	Description	Indicator
Letter Name Knowledge	Measure automaticity in letter recognition	A grid of 100 letters that students choose from to identify as many letters as they can (1 minute)	Letters read per minute
Letter Sound Knowledge	Measure automaticity in knowledge of the sounds associated with each letter	A grid of 100 letters that students use to sound out as many letters as they can (1 minute)	Letter sounds pronounced per minute
Familiar Word Reading	Measure skill in reading high-frequency words	A grid of 50 words that students use to read as many words as they can (1 minute)	Words read per minute
Invented Word Decoding	Measure skill in applying letter-sound correspondence rules to decode (i.e., sound out) unfamiliar words	A grid of 50 unfamiliar words that students use to read as many words as they can (1 minute)	Invented words read per minute
Oral Reading Fluency	Measure fluency in reading a passage of grade-level text aloud	A short passage that students read aloud (1 minute)	Words in a story read per minute
Initial Sound Identification	Measure phonemic awareness (ability to explicitly identify and manipulate the sounds of language)	Ten words are read by the tester, and students identify the sound of the first letter of the word (untimed).	Sounds correctly identified (percentage)
Reading Comprehension	Measure comprehension of text read in the Oral Reading Fluency subtask	Questions about the text that students read (maximum 5; untimed)	Questions correctly answered (percentage)
Listening Comprehension	Measure ability to comprehend a story presented orally	5 questions about the text read by the tester (untimed)	Creole and French oral comprehension (percentage)

Note. See Dubeck and Gove (2015).

Statistical Methods

Attrition

During the two follow-ups, some schools and students were not able to be recontacted. To measure the impact of attrition at the end of grades 1 and 2, we adopted the guidelines outlined by the What Works Clearinghouse (WWC; 2017c) to examine attrition for clustered randomized controlled trial designs. The WWC protocol establishes three levels to assess the quality of studies: (1) meet the WWC group design standards without reservations, (2) meet the WWC group design standards with reservations, and (3) do not meet the WWC group design standards. To meet the group design standards without reservations, there should not be a risk of bias due to attrition. To meet the group design standards with reservations, there should be equivalence of individuals at baseline for the groups in the analytical sample if there is a risk of bias due to attrition.

Establishing the Risk of Bias Due to Attrition

In this study, schools were assigned to treatment and control groups, and we analyzed individual outcomes within each school to determine the impact of Read to Learn. Therefore, attrition levels had to be considered at two levels: the cluster level (school) and the individual level within clusters (students). The risk of attrition bias is determined by examining the combination of overall and differential attrition (WWC, 2017c) at the school and individual levels. Overall attrition is calculated by the percentage of schools or individuals remaining in the sample of the initial number of schools or individuals, respectively. Differential attrition is the percentage point difference in the attrition rates calculated for the treatment and control groups.

Equivalence of Individuals at Baseline

When the combination of overall attrition and differential attrition are above the WWC standard threshold, then the equivalence of individuals at baseline needs to be examined. First, equivalence must be established by comparing the characteristics measured at baseline with those of the sample of students retested at each follow-up. Each characteristic is measured in terms of effect size. The recommended effect size for continuous variables is Hedges's g , and for binary outcomes, it is the Cox index (WWC, 2017b). When the effect size is greater than 0.25, the WWC considers the intervention and control groups to be nonequivalent. For effect sizes between 0.05 and 0.25, the analysis must include an acceptable statistical adjustment for baseline characteristics. In this article, we include such characteristics in the statistical model. For effect sizes less than or equal to 0.05, the WWC does not require a statistical adjustment.

Statistical Model

To account for differences in the sample sizes at grades 1 and 2 and the equivalence tests and outcome variables available at baseline, we conducted the analyses of the two grades separately, and we used two methods to measure the impact of Read to Learn. We used a hierarchical linear model approach to account for students being clustered within schools. The effect of the intervention was measured using an analysis of covariance model. This model required that we regress the outcome variable at the end of grade 1 or 2 on a treatment indicator variable, the baseline value of the outcome variable, and the other variables identified in the equivalence tests. The analysis of covariance model with covariates provides more precision and statistical power to detect differences in evaluation designs (Bloom, Richburg-Hayes, & Black, 2007; Raudenbush & Bryk, 2002). The final estimation model is as follows:

$$y_{ijk} = \delta_0 + \gamma T_j + \delta_1 y_{ij0} + \delta_2 \mathbf{X}_{ij0} + v_j + \omega_{ij} \quad (1)$$

where y_{ijk} is the outcome variable at k , the end of grade 1 or 2 of student i in school j . The parameter of interest is γ , which measures the treatment effect, as T is an indicator variable that equals 1 for treatment schools and 0 for control schools. The baseline and its coefficient are represented by $\delta_1 y_{ij0}$. The $\delta_2 \mathbf{X}_{ij0}$ portion is boldfaced to indicate that it is not just one coefficient and one variable; it represents all variables measured at baseline identified in the equivalence analysis and their corresponding coefficients. The between-school variation is measured by v_j , and the within-school variation is captured by ω_{ij} ; they have a mean of 0, with variance σ_0 and σ_1 to be estimated and zero covariance.

We used the baseline value of the outcome for all of the reading skills except reading comprehension and listening comprehension. At baseline, the server used for Tangerine failed to properly collect the data from the tablets. We were able to recover the results for all of the subtests except reading comprehension and listening comprehension.¹ For these sections of the test, the results do not include the $\delta_1 y_{ij0}$ term. The omission of this variable does not affect the estimation of the effect size but could reduce the statistical power to detect the effect of the intervention.

Effect Sizes

We measured the magnitude of the impact of Read to Learn with effect sizes. For all the EGRA outcome measures, we used Hedges's g (WWC, 2017b), which is computed as the covariate-adjusted mean difference divided by the unadjusted pooled within-group standard deviation (for details and formulas, see WWC, 2017b).

Results

Participant Flow

At the beginning of first grade, 1,189 students were tested in 94 schools. At the end of first grade, 930 students in 88 schools were retested, and at the end of second grade, 634 students in 92 schools were retested (see Table 3). Data were collected at the beginning of first grade between October 6 and 10, 2014. Between May 27 and June 2, 2015, test administrators tested students at the end of the first-grade school year. At the end of the second-grade school year, student data were collected between May 20 and June 8, 2016.

Attrition

Attrition at the School Level

At the end of first grade, the combined overall and differential attrition was well within the limits of the WWC standards of minimal risk of attrition bias (WWC, 2017a). We obtained data from 88 of the 94 schools that we randomly assigned to the treatment and control groups (see Table 3). This resulted in a 6% overall attrition rate at the cluster level. In the treatment group, 42 of 45 schools remained in the sample (6.7% attrition), and in the control group, 46 of 49 schools remained in the sample (6.1% attrition), with a resulting differential attrition level of 0.5%.

At the end of second grade, there was also minimal risk of attrition bias. We were able to return to schools that had not been visited at the end of first grade. Therefore, the number of schools that were revisited increased to 92 of 94, resulting in an overall level of attrition of 2%. In the treatment group, 43 of 45 schools were revisited (4.4% attrition). There was no attrition in the control group. Hence, the differential attrition was 4.4%.

Attrition at the Individual Level

We measured individual-level attrition among the students in nonattriting schools at the end of grades 1 and 2, following the WWC (2017c) guidelines. The attrition levels and risk of attrition bias varied for the two grades.

At the end of first grade, the combined overall and differential attrition levels did not present a risk of attrition bias. Eighty-eight schools were revisited at the end of

second grade. The total number of students from these 88 schools who were tested at baseline was 1,143. A total of 930 of these students were retested, for an overall attrition rate of 19%. In the treatment group, 444 of 545 students at nonattriting schools who were tested at baseline were retested in the treatment group (18.7% attrition). In the control group, 486 of 598 students at nonattriting schools who were tested at baseline were retested (18.5% attrition). The differential attrition was 0.2%.

At the end of second grade, the overall and differential attrition indicated that there was a risk of attrition bias. Ninety-two schools were revisited at the end of second grade. The total number of students from these 92 schools who were tested at baseline was 1,184. A total of 634 of these students were retested at the end of second grade (46% attrition). The number of students retested in the treatment schools was 276 of 552 (50% attrition). The number of students retested in the control schools was 383 of 632 (43.4% attrition). Thus, the differential attrition was 6.6 percentage points between the treatment and control schools.

Equivalence Tests

Next, we examined whether the samples of students in the treatment and control groups were still equivalent to measure the impact of the program after considering attrition. If the composition of the treatment and control groups was different, then the impact of the study might be attributed to those differences, not to the effectiveness of the program. Thus, we compared the average student characteristics of the treatment and control groups. These characteristics are best evaluated at baseline. We collected students' age and sex, as any other socioeconomic or background characteristics could not reliably be obtained from the students or from records kept at the Haitian schools. We also examined the two regions, the average first-grade class size, and all of the outcome variables measured at baseline.

At the end of first grade, the values of six variables were over the 0.05 threshold but under the 0.25 threshold (see Table 4). Consequently, the analyses required statistical correction through the use of sex, average class size, region, letters read per minute, letter sounds pronounced per minute, and words read per minute as controls. At the end of second grade, the equivalence tests indicated that

TABLE 3
Number of Schools and Students Tested in Each Round of Data Collection

Grade	Schools			Students		
	Total	Control	Treatment	Total	Control	Treatment
Baseline	94	49	45	1,189	632	557
End of grade 1	88	46	42	930	486	444
End of grade 2	92	49	43	634	358	276

TABLE 4
Absolute Effect Sizes at Baseline for the Samples Tested at the End of Grades 1 and 2

Variable	End of grade 1					End of grade 2				
	Control (n = 486)		Treatment (n = 444)		Effect size	Control (n = 358)		Treatment (n = 276)		Effect size
	M	SD	M	SD		M	SD	M	SD	
Student characteristics										
Sex (percentage female) ^a	53.3	50.0	52.7	50.0	0.058	53.1	50.0	52.2	50.0	0.022
Age (years)	5.5	1.1	5.5	1.1	0.047	5.6	1.2	5.6	1.2	0.006
School characteristics										
Average first-grade class size	38.2	12.8	37.0	12.5	0.091	38.4	12.5	36.7	12.7	0.157
Region (percentage in Gonaïves) ^a	54.0	49.9	62.2	48.5	0.205	58.9	49.3	62.3	48.5	0.086
Early Grade Reading Assessment subtasks										
Letters read per minute	8.7	13.5	9.6	13.4	0.065	10.5	14.7	10.7	13.0	0.017
Letter sounds pronounced per minute	7.4	11.1	8.5	15.0	0.077	8.5	12.2	10.3	16.7	0.125
Words read per minute	3.5	7.7	4.1	14.3	0.052	4.2	9.0	4.3	14.3	0.005
Invented words read per minute	3.0	8.5	3.0	10.0	0.001	3.9	10.0	3.4	10.3	0.048
Words in a story read per minute	6.6	17.0	7.2	22.1	0.027	8.4	19.9	6.0	16.1	0.132
Initial sound recognition	18.9	31.5	20.5	33.2	0.048	21.8	32.8	23.9	35.8	0.060

Note. M = mean; SD = standard deviation. The Hedges's g formula was used to calculate the effect sizes.

^aThe Cox index was used for the categorical variables.

the analyses needed to control for five different variables: average class size, region, letter sounds pronounced per minute, words read in a story per minute, and initial sounds recognized. In addition, we included sex as a control because it showed a strong correlation in the results for first grade, and this correlation was also confirmed in the results for the end of second grade.

Regression Model Results

Read to Learn had positive and statistically significant effects on students for most of the reading skills measured at the end of grades 1 and 2. Table 5 presents the regression coefficients for both grades and all control variables. Table 6 presents the effect sizes and the components used for their calculations.

At the end of first grade, the differences between the treatment and control schools in six of the eight skills measured on the EGRA were statistically significant at $p < .05$. These skills were letters read per minute ($p = .020$, effect size [ES] = 0.32), letter sounds pronounced per minute ($p < .001$, ES = 0.47), words read

per minute ($p = .016$, ES = 0.43), invented words read per minute ($p = .035$, ES = 0.19), words in a story read per minute ($p = .026$, ES = 0.20), and initial letter sound recognition ($p < .001$, ES = 0.77). At the end of second grade, the differences between the treatment and control groups in six of the eight skills were statistically significant. These skills were letters read per minute ($p = .003$, ES = 0.35), letter sounds pronounced per minute ($p < .001$, ES = 0.79), words read per minute ($p = .001$, ES = 0.35), invented words read per minute ($p = .004$, ES = 0.32), words in a story read per minute ($p = .002$, ES = 0.34), and initial sound recognition ($p < .001$, ES = 0.79).

Regarding the variables included as a result of the equivalence tests, the results in the regression show that sex and region were statistically significant across most of the subtasks. Sex was statistically significant in six of the nine outcomes at the end of first grade and seven of the nine outcomes at the end of second grade. In all of these cases, females had higher scores than males. Region was statistically significant for listening comprehension at the end of first grade and for five of the nine

TABLE 5
Regression Coefficients Measuring the Impact of the Read to Learn Program at the End of Grades 1 and 2 in Haiti

Variable	Letters	Letter sounds	Words	Invented words	Story	Initial sound	Comprehension		
							Reading	Oral Creole	Oral French
End of grade 1 (n = 930)									
Treatment	5.90* (1.88)	7.63* (1.98)	3.20* (1.37)	2.26* (1.07)	4.11* (1.85)	29.79* (4.43)	6.51 (2.97)	5.89 (3.87)	8.71 (4.71)
Average class size	-0.04 (0.07)	0.06 (0.07)	0.02 (0.05)	-0.01 (0.04)	0.02 (0.07)	0.04 (0.17)	-0.01 (0.11)	-0.15 (0.15)	-0.02 (0.18)
Sex	4.47* (0.99)	4.24* (0.92)	3.62* (0.80)	3.01* (0.67)	5.50* (1.22)	7.45* (2.22)	2.26 (1.96)	-0.01 (1.77)	1.63 (1.48)
Region	0.19 (1.91)	-1.75 (2.00)	-0.04 (1.39)	0.02 (1.08)	1.19 (1.87)	-7.29 (4.49)	4.83 (3.01)	-11.70* (3.93)	-9.45* (4.79)
Baseline score	1.31* (0.23)	0.39 (0.34)	0.21 (0.29)	-1.59* (0.35)	-0.07 (0.14)	0.51* (0.16)			
Letters read per minute baseline		0.34 (0.24)	0.75* (0.17)	0.69* (0.13)	0.96* (0.23)	-0.02 (0.55)	1.26* (0.37)	0.61 (0.47)	1.67* (0.57)
Letter sounds read per minute baseline	-0.36 (0.32)		-0.38 (0.23)	-0.24 (0.18)	-0.39 (0.31)	0.29 (0.79)	-0.21 (0.50)	-0.64 (0.66)	-0.52 (0.81)
Words read in a story per minute baseline	-0.07 (0.40)	-0.52 (0.42)		1.39* (0.33)	0.60 (0.41)	-0.33 (0.95)	0.23 (0.62)	0.32 (0.82)	-0.84 (1.00)
Constant	19.58* (3.23)	10.54* (3.38)	7.23* (2.36)	5.92* (1.86)	8.54* (3.20)	34.23* (7.58)	12.61* (5.16)	59.77* (6.61)	26.15* (7.98)
Mean variance	1.99* (0.11)	2.08* (0.10)	1.62* (0.12)	1.34* (0.12)	1.85* (0.13)	2.86* (0.11)	2.32* (0.13)	2.76* (0.10)	3.02* (0.08)
Residual variance	2.68* (0.02)	2.59* (0.02)	2.46* (0.02)	2.28* (0.02)	2.89* (0.02)	3.48* (0.02)	3.37* (0.02)	3.25* (0.02)	3.07* (0.02)

(continued)

TABLE 5
Regression Coefficients Measuring the Impact of the Read to Learn Program at the End of Grades 1 and 2 in Haiti (continued)

Variable	Letters	Letter sounds	Words	Invented words	Story	Initial sound	Comprehension		
							Reading	Oral Creole	Oral French
End of grade 2 (n = 634)									
Treatment	8.11* (2.77)	18.93* (2.94)	8.98* (2.71)	6.34* (2.23)	11.69* (3.77)	30.54* (4.95)	7.48 (4.56)	4.09 (3.33)	7.69 (4.66)
Average class size	0.03 (0.11)	0.17 (0.12)	0.15 (0.11)	0.12 (0.09)	0.18 (0.15)	-0.02 (0.19)	-0.10 (0.18)	-0.21 (0.13)	-0.16 (0.18)
Region	-6.12* (2.82)	-7.95* (2.99)	-2.16 (2.74)	-2.92 (2.23)	-4.85 (3.82)	-14.81* (5.04)	-3.16 (4.63)	-11.95* (3.39)	-20.82* (4.75)
Sex	3.54* (1.57)	2.94 (1.69)	5.61* (1.80)	4.62* (1.45)	10.88* (2.46)	2.72 (2.75)	5.70* (2.80)	5.18* (2.09)	6.00* (2.07)
Baseline score	0.79* (0.33)	-0.35 (0.24)	0.57 (0.63)	0.02 (0.47)	-0.36 (0.25)	0.11 (0.17)			
Letter sounds pronounced per minute baseline	-0.53 (0.36)		0.03 (0.40)	0.24 (0.27)	0.57 (0.31)	-0.04 (0.40)	0.08 (0.37)	-0.81* (0.27)	-0.36 (0.38)
Words in a story read per minute baseline	-0.06 (0.18)	-0.02 (0.20)	-0.25 (0.19)	-0.13 (0.17)		0.30 (0.33)	-0.04 (0.30)	0.50* (0.22)	-0.05 (0.31)
Initial sound recognition baseline	0.27* (0.09)	0.29* (0.10)	0.28* (0.09)	0.25* (0.07)	0.41* (0.13)		0.66* (0.15)	0.23* (0.11)	0.34* (0.16)
Constant	37.37* (4.93)	18.13* (5.24)	17.10* (4.88)	13.49* (3.97)	24.35* (6.75)	56.21* (8.80)	42.58* (8.14)	78.52* (5.96)	50.49* (8.20)
Mean variance	2.36* (0.11)	2.42* (0.11)	2.23* (0.13)	2.03* (0.13)	2.58* (0.13)	2.95* (0.11)	2.81* (0.12)	2.49* (0.13)	2.97* (0.09)
Residual variance	2.93* (0.03)	3.00* (0.03)	3.07* (0.03)	2.86* (0.03)	3.38* (0.03)	3.48* (0.03)	3.51* (0.03)	3.22* (0.03)	3.19* (0.03)

Note. Standard errors are in parentheses.

* $p < .05$.

TABLE 6
Effect Sizes on the Impact of the Read to Learn Program in Haiti

Early Grade Reading Assessment subtask	Unadjusted mean		Unadjusted standard deviation		Covariate-adjusted mean		Effect size
	Treatment	Control	Treatment	Control	Treatment	Control	
End of grade 1 (treatment n = 444; control n = 486)							
Letters read per minute	25.8	20.3	19.0	17.4	25.7	19.8	0.32
Letter sounds pronounced per minute	22.3	14.2	18.4	14.2	21.5	13.8	0.47
Words read per minute	13.0	9.8	14.9	12.9	12.6	9.4	0.23
Invented words read per minute	9.9	7.2	12.5	10.6	9.4	7.1	0.19
Words in a story read per minute	16.4	12.4	21.7	19.9	16.1	12.0	0.20
Initial sound recognition	66.2	36.2	38.8	38.6	65.3	35.5	0.77
Reading comprehension	23.3	15.8	34.3	31.0	22.0	15.5	0.20
Oral Creole comprehension	54.0	48.5	30.7	31.0	53.0	47.1	0.19
Oral French comprehension	30.7	21.4	33.3	28.7	29.3	20.6	0.28
End of grade 2 (treatment n = 276; control n = 358)							
Letters read per minute	44.9	37.4	24.6	21.4	45.0	36.9	0.35
Letter sounds pronounced per minute	40.2	22.3	28.1	20.1	40.5	21.6	0.79
Words read per minute	33.5	24.9	26.8	22.7	33.3	24.6	0.35
Invented words read per minute	25.5	19.1	21.9	18.4	25.2	18.9	0.32
Words in a story read per minute	46.8	34.8	36.8	32.4	46.1	34.4	0.34
Initial sound recognition	78.6	48.1	34.0	42.1	78.5	48.0	0.79
Reading comprehension	52.5	44.0	30.7	39.5	51.9	44.5	0.19
Oral Creole comprehension	70.1	67.2	28.6	29.4	70.4	66.3	0.14
Oral French comprehension	43.8	35.0	34.7	31.8	43.3	35.6	0.23

Note. The Hedges's *g* formula was used to calculate the effect sizes using the covariate-adjusted mean and the unadjusted standard deviation (WWC, 2017b).

subtasks at the end of second grade. The statistically significant coefficients for region were negative. This result indicates that students from Cap Haitian outperformed students from Gonaïves.

In summary, at the end of grades 1 and 2, Read to Learn had impacts on all of the skills related to letter name and sound knowledge, sound recognition, decoding, and oral reading fluency. However, no impact was detected on reading or listening comprehension.

Discussion

The research question for this study was whether the Read to Learn program had an impact on the literacy skills of Haitian students. Our results indicate that Read to Learn had an impact on most reading skills. These

findings are notable in that they add to the small but growing body of evidence that systematic literacy and training programs can be effective in improving students' literacy skills (Evans & Popova, 2015; Kim et al., 2016; Macdonald & Vu, 2018; Snilstveit et al., 2016). There is limited peer-reviewed research on the impact of scripted early-grade literacy programs in fragile contexts, but there is promising evidence from project reports that aligns with our results. A study in Liberia reported that a targeted, scripted reading intervention focused on improving the quality of reading instruction for primary school students had a large impact on student literacy skills in a relatively limited amount of time (Piper & Korda, 2011). In a program in Nigeria, teachers implemented effective instructional practices, such as reading aloud and frequently asking comprehension questions to students, which led descriptively to an

increase in the overall distribution of reading comprehension scores but did not achieve statistically significant results (RTI International, 2016). A scripted literacy program in Uganda and Kenya found positive effects in both contexts but were larger in Uganda than Kenya, which the authors tentatively attributed to variations in country-specific language instruction (Lucas et al., 2014).

The positive impact of Read to Learn on literacy skills through the use of a mother-tongue curriculum confirms the benefits of a focus on mother-tongue literacy approaches in multilingual contexts more generally (Ball, 2010; Brunette, Piper, Jordan, King, & Nabacwa, 2019; Kosonen & Young, 2009; Piper, Schroeder, & Trudell, 2016). Our results also support a focus on Haitian Creole for literacy acquisition for young students in Haiti (DeGraff, 2016). Our results also align with findings from Kenya, in which researchers compared two treatment arms of a scripted literacy program, one in English and Kiswahili (a common second language) and one in English, Kiswahili, and students' mother tongue. The researchers found larger and more consistent program effects on the mother-tongue intervention than the other treatment arm that did not include mother-tongue instruction (Piper, Zuilkowski, & Ong'ele, 2016). In Uganda, a scripted curriculum program had an impact on students' literacy achievement in nine of 12 languages in which the program was implemented (Brunette et al., 2019).

The impact of Read to Learn on letter name and sound knowledge, decoding, and sound recognition bodes well for participating students' future literacy gains. Studies across various contexts have linked these gains in emergent literacy skills to reading comprehension and future literacy achievement (Sabatini, Wang, & O'Reilly, 2019). A study in Kenya showed the positive effects of receiving basic literacy interventions in early grades through the Primary Math and Reading Initiative, particularly in low-income contexts (Piper, Jepkemei, & Kibukho, 2015). Students in Costa Rica who received high-quality literacy interventions focused on phonological awareness in early grades were able to reduce future incidences of reading difficulties (San Francisco, Arias, Villers, & Snow, 2006).

Our Read to Learn findings provide evidence that scripted literacy programs can improve reading fluency. There is limited evidence available on this topic in developing countries, but the effect of Read to Learn is consistent with those studies that exist. A study in Ethiopia of a scripted literacy program demonstrated that working with teachers, parents, and community members, as well as providing strong reading materials, boosted reading fluency gains (Friedlander et al., 2012). In Bangladesh, similar results in reading fluency were achieved with a scripted program, and the researchers suggested that students with higher results in reading fluency could serve as

valuable resources for their nonreading peers to improve their performance (Jonason, Guajardo, Nath, & Hossain, 2014). Achieving statistically significant gains in reading fluency in Haiti is particularly notable, as no previous intervention has demonstrated gains in oral reading fluency in Haiti (RTI International, 2015).

The impact of Read to Learn on oral reading fluency is particularly important because of the strong relation between reading fluency and reading comprehension. Research has shown that fluency in foundational skills, including grade-level decoding and word recognition, are strongly associated with prosody and reading comprehension (Sabatini et al., 2019). Oral reading fluency predicts both current and future (longer term) reading comprehension (Klauda & Guthrie, 2008; Roehrig, Petscher, Nettles, Hudson, & Torgesen, 2008; Valencia et al., 2010). A number of studies have explored the empirical and theoretical links between oral reading fluency and reading competence (see, e.g., Eason, Sabatini, Goldberg, Bruce, & Cutting, 2013; García & Cain, 2014; Kim, Wagner, & Foster, 2011; Kim, Wagner, & Lopez, 2012; Silverman, Speece, Harrington, & Ritchey, 2013; Tilstra, McMaster, Van den Broek, Kendeou, & Rapp, 2009). Research has suggested that in more transparent orthographies such as Creole, oral reading fluency plays an even larger role in reading comprehension than in reading accuracy (Cadime et al., 2017; Padeliadu & Antoniou, 2014; Santos, Cadime, Viana, & Ribeiro, 2019; Seguin, 2018). When students are able to read words with accuracy and speed, cognitive constraints are lifted, "allowing cognitive resources (e.g., working memory, attention) to be used for higher order meaning construction" (Kim, 2015, p. 460) that is needed for reading comprehension. Considering all of this evidence, we are optimistic that the gains achieved in Creole oral reading fluency will equip students who participated in Read to Learn to make substantial future gains in reading comprehension.

The results of Read to Learn provide evidence about what could work to improve literacy outcomes for young students in other resource-scarce, multilingual contexts with low overall literacy rates. Haiti, along with 46 other countries, is classified as a least developed country (LDC) that faces severe structural barriers to human development (United Nations Conference on Trade and Development [UNCTAD], 2019). The 47 identified LDCs are home to approximately 880 million people and 12% of the world population (UNCTAD, 2019). The average literacy rates of 15–24-year-olds in LDCs is 73.8% (UNCTAD, 2018). In 44 of the 47 LDCs, multiple languages are spoken by a considerable portion of the population, meaning that literacy instruction has to take into account a variety of mother tongues and, often, the use of a lingua franca in the education system (Central Intelligence Agency, 2019). Other LDCs like Haiti could benefit from the approach and valuable lessons presented by the results of Read to Learn.

Interpretation of Results

Despite the causal evidence presented and potential applicability of our results in other contexts, it was difficult to determine why Read to Learn worked or failed. Randomized controlled trials are very good at indicating whether a treatment has an average impact on an outcome. However, they are very limited in revealing why the treatment worked (or not). It was not possible to determine, within the scope of this study, which components of the intervention explain the results of this program.

The lack of reliable process data on key indicators also limited our ability to acquire a full understanding of which elements of Read to Learn worked. It would be helpful, in contexts of scarce resources like Haiti, to know which elements of the program were most effective to focus limited funds on those practices. This information could be used to conduct mediation and moderation analysis of treatment effects (Emsley, Dunn, & White, 2019; Kelcey, Dong, Spybrook, & Cox, 2017). Potentially important elements to explore include the percentage of total lessons completed per teacher per year in both languages, teacher attendance at trainings, the number of observation and coaching visits received per teacher, and the availability of adequate project materials for all students and teachers. Additionally, indicators related to instructional time, such as the number and reasons for lost instructional days and student and teacher attendance, are valuable.

We examined some data and reports from the instructional coaches. We found that over 92% of teachers had the teaching guides and that there was high teacher attendance at trainings. The numbers of observation and coaching visits were also higher than in previous similar projects in Haiti (RTI International, 2015). Coaches were supposed to collect data on tablets and upload them to a central cloud-based system. However, these data from observation visits were not regular or complete. A lack of electricity and internet connectivity hindered this process. There are inevitable challenges in conducting research in contexts such as Haiti, ranging from a lack of trained researchers to a lack of enumerators capable of following research protocols (Kim et al., 2016).

This study also showcases the main problems inherent in a longitudinal investigation in the field. Although there was no large attrition at the school level, there was at the student level. There was also differential student attrition between the treatment and control schools. Following the guidelines for longitudinal analysis set by the WWC (2017c), the attrition level for the first-grade results led to minimal risk of bias due to attrition. The attrition level for second grade, however, raises some concerns. The equivalence test showed that the size of the differences between the treatment and control schools could be dealt with statistically, so we included variables in the estimations to

control for those differences. Following the WWC guidelines, the data for first grade are eligible to meet WWC group standards without reservations, whereas for second grade, the data are eligible to meet WWC group standards with reservations. It is encouraging to see consistent effect sizes and coefficients across results for both grades. We believe that this reaffirms our conclusion that the results for second grade are not upwardly biased, which could occur if there was attrition of high performers in control schools or low performers in treatment schools. In sum, we concluded that the internal validity of the results was not threatened by attrition, given that it was not a problem for first grade and that we properly accounted for any nonequivalence in second grade.

Two of the variables that were not equivalent between the treatment and control groups were statistically significant in the multivariate analysis: region and sex. In addition, letters read per minute was statistically significant for the second-grade results. The inclusion of nonequivalent variables should control for differences in the sample composition. Other techniques to balance the two samples are available, such as matching or weighting. However, these techniques affect the sample size or statistical power without guaranteeing a reduction in bias.

Future research designs need to recognize that in developing countries, student tracking and dropouts are a threat to longitudinal studies. One clear effect of attrition in the current study was the reduction in statistical power. In future research in settings such as Haiti, high levels of attrition should be expected, and oversampling at baseline is a solution in these contexts. We underestimated how difficult it would be to follow students over time. Lack of student information and students dropping out are significant challenges. Poor information stems from unreliable student rosters and an education system that does not track student transfers.

There are some control mechanisms that could aid in minimizing levels of attrition in countries like Haiti: pictures of student rosters so students can be effectively tracked for repeated testing, engaging school staff to aid in the identification of students and understanding student inflow and outflows from schools, and adding extra schools into the initial study design to take into account schools that will not be accessible for testing due to weather, road conditions, and school closures to prevent losing schools due to these events.

The assessment that we used is another limitation of this research. Our study relied only on the EGRA. Some have argued that the EGRA focuses too much on fluency across subtasks and has a problematic reading comprehension measure (Bartlett et al., 2015; Hoffman, 2012). Indeed, reading fluency is necessary but not sufficient for assessing gains in reading comprehension (Klauda & Guthrie, 2008). Future work should include assessments

that are more sensitive to reading comprehension skills, such as vocabulary acquisition.

Conclusion

Even with the limitations outlined earlier, the results of this study are an important contribution to knowledge about what works to improve literacy outcomes for students in Haiti and other fragile contexts. Despite many difficulties, the results demonstrate considerable gains in most literacy skill areas. These results represent the strongest results to date in Haiti and offer solid justification for a broader replication of this approach in more Haitian schools. A program such as Read to Learn might be a solution to the learning crisis in additional poor and fragile contexts.

Future research should focus on several key areas. In the current study, we did not explore variation in the impact of the program among different groups, nor did we examine mediation and moderation effects. Future research could focus on both of these areas. Multifactor randomized controlled trials can be used to test which program components are most cost-effective. Resources should be allocated to collecting classroom-level observational data on teacher practice and implementation fidelity. Progress in these areas will contribute to determining the most cost-effective and scalable suite of interventions capable of transforming the life outcomes of students in fragile contexts.

NOTES

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¹In our field experience in Haiti and in other contexts, we learned that Tangerine, the application that must be used to collect EGRA data, required a certain level of internet speed to upload results. The data collected during baseline were corrupted on the server with repeated attempts to upload the data with a subpar internet connection. The data were partially recovered directly from the tablets, but the data on reading and listening comprehension were not recovered.

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JUAN CARLOS GUZMÁN (corresponding author) was a senior researcher at the Pulte Institute for Global Development at the University of Notre Dame, Indiana, USA, at the time of this study; email jkguzman1@gmail.com. His research focuses on impact evaluation of projects in education, health, and labor markets in developing countries.

KATE SCHUENKE-LUCIEN is the director of Haiti Initiatives at the Alliance for Catholic Education at the University of Notre Dame, Indiana, USA; email kschuenk@nd.edu. Her research focuses on improving life outcomes for vulnerable children and families globally.

ANTHONY J. D'AGOSTINO is faculty program director of the International Education Research Initiative and an assistant professor in the Institute for Educational Initiatives at the University of Notre Dame, Indiana, USA; email adagosti@nd.edu. His scholarship focuses on comparative education policy, faith-based schools, school effectiveness, and early-grade literacy and systems improvement in developing country contexts.

MARK BERENDS is the director of the Center for Research on Educational Opportunity and a professor in the Department of Sociology at the University of Notre Dame, Indiana, USA; email mberends@nd.edu. His research interest include inequality, opportunity gaps, and school choice.

ANDREW J. ELLIOT is a professor in the Department of Psychology at the University of Rochester, New York, USA; email andye@psych.rochester.edu. He conducts research on approach and avoidance motivation, especially in achievement contexts such as school, sports, and work.